Experiment - 7

Flow control at Data Link layer using Sliding Window protocol

Aim:

To write a program to implement flow control at the Data Link Layer using the sliding Window protocol.

Algorithm:

1) Start the program
2) Read the total number of frames (N), window size (W), & the frame number to be lost.
3) Initialize :
   - base = 0 - first unacknowledged frame
   - nextSeq = 0 - next frame to be sent
4) While base < N, determine the sender window from base to base + W - 1
5) Send all frames within the current window and increment nextSeq after sending each frame.
6) If the specified frame is lost.
   - Receiver detect the missing frame
   - Receiver discards the out-of-orders frames.

* Sender experience a timeout
* Sender retransmit the frames starting from the lost frame.

7) If there is no frame loss:
* Reciver accepts the frames in order
* Recives sends an ACK for each recived frame
* Sender reciver the ACK and slides the window forward.
* Set base = nextSeq

8) Repeat steps 4-7 untill all frames are successfully transmitted & acknowleged

9) Display that all frames

10) Stop the program

Code:-

```
import java.util.Scanner;
public class SlidingWindow {
    static void simulate(int totalFrames,
        int windowSize, Integer lostFrame) {
        int base = 0;
        int nextSeq = 0;
        while (base < totalFrames) {
            int windowEnd = Math.min(
                base + windowSize - 1,
                totalFrames -
            );
```

```
System.out.println("Sender Window: [" + base
    + " - " + windowEnd + "]");

while (nextseq < base + windowSize && nextseq <
        totalFrames) {
    System.out.println("Node A -> Sending Frame"
        + nextseq);

    if (lostFrame != null && nextseq == lostFrame) {
        System.out.println("Frame" + nextseq +
            "Lost in transit)");
        );
    }
    nextseq++;
}

if (lostFrame != null && base <= lostFrame
    && lostFrame < nextseq) {

    System.out.println(
        "Node B -> Frame" + lostFrame +
        "missing"
    );

    System.out.println(
        "Node A -> Timeout! Resending
        window starting from Frame" +
        lostFrame + ".");

    nextseq = lostFrame;
```

```
            lostFrame = null;
            continue;
        }

        for (int f = base; f < nextSeq; f++) {
            System.out.println(
                "Node B -> Reived Frame " + f + ", sending ACK " + f
            );
        }
    );

    base = nextSeq;
    }
);
}

public static void main(String[] args){
    Scanner sc = new Scanner(System.in);
    System.out.println(
        "Enter total number of frames to send : ");
    int totalFrames = sc.nextInt();
    System.out.println(
        "Enter sender's window size : ");
    int windowSize = sc.nextInt();
    System.out.println(
        "Enter frame numbers to simulate as LOST\n" +
        "(0 to " + (totalFrames - 1) + ", enter
            -1 for no loss): "
    );
```

```
int inputLostFrame = sc.nextInt();
    Integer lostFrame;
if (inputLostFrame == -1){
        lostFrame = null;
}else {
    lostFrame = inputLostFrame;
  }
simulate (totalFrames, windowSize,
                                lostFrame);
  sc.close();
 }
}
```

Output:-

```
Enter total number of frame to send : 3
Enter sender's window size : 2
Enter frame number to simulate as LOST
  (0 to 2, enter -1 for no loss) : 1

Sender window : [Frame 0 - Frame 1]
  Node A -> Sending Frame 0
  Node B -> Sending Frame 1
            (Frame 1 LOST in transmit)
  Node B -> Frame 1 missing
  Node A -> Timeout ! Resending from Frame 1

Sender Window : [Frame 1 - Frame 2]
  Node A -> Sending Frame 1
  Node A -> Sending Frame 2
```

Node B -> Reived Frame 1, sending Ack 1

Node B -> Reived Frame 2, sending Ack 2

Node A -> Ack 2 reived. Sliding window forward.

All 3 frames successfully transmitted from Node A to Node B

Result:- Thus the program to implement flow control at the Data Link layer using the Sliding window protocol was implemented successfully.